

Joanne Hoar

Technology Educator, Software/Game Developer



linkedin: Joanne Hoar

github: xcodegirl

WHO AM I?

Canadian Software Developer with 20 years of experience in video game programming, scientific modelling and cross-platform development. Expertise in multiple programming languages, game engines and SDKs. Proven track record in leading teams, mentoring developers, impressing customers and building corporate culture. Strong background in C++, Unreal Engine, iOS and Android using agile methodologies and CI/CD. Passion for algorithmic correctness, mathematics, artificial intelligence and nature.

EXPERIENCE

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|--------------------|---|
| Jan 2025 – present | Technology Instructor
Robertson College (contract)
Instructed 20+ course offerings across Full Stack Development and Data Analytics diploma programs. Delivered live coding sessions, maintained open-source course materials on GitHub, and achieved an average student satisfaction rating of 4.5/5.0.
C# / .NET / ASP .NET Core / Java / JavaScript / React / Git / GitHub / Azure / SQL |
| 2022 – 2025 | Senior Software Developer
bitHeads Inc.
Developed and maintained cross-platform client SDKs for a backend-as-a-service platform used by 50+ published games across 12 platforms including PlayStation, Xbox, Nintendo Switch, iOS, and Android. Engineered CI/CD and TDD pipelines and managed customer relationships with AAA and Indie studios.
Unreal Engine / C++ / Objective-C / Swift / iOS / Android Native / Java / Kotlin / PlayStation / Xbox / Nintendo Switch / Node.js / Groovy / C# / Shell Script |
| 2011 – 2022 | Co-founder/ Lead Developer
Silicon Hanna Inc. (self-employed)
Designed, coded and published eleven mobile games for iOS and Android, from concept and game design through to commercial app-store release. Experimented with cutting-edge device technologies including ARKit and other emerging SDKs.
Unreal Engine / Objective-C / Swift / iOS / Android Native / Kotlin / Augmented Reality / MacOS / Windows |
| 2008 – 2011 | Tools Programmer
Fekete Associates Inc. (S&P Global)
Developed tools that used mathematical modelling to maximize production and improve efficiency for major oil & gas corporations. Led research on a new system that reduced build times by 40%.
C++ / MFC / Windows |
| 2005 – 2008 | Intermediate Software Engineer
BJ Pipeline Inspection Services Inc. (Baker Hughes)
Visualised in-line device data with OpenGL graphics. Implemented neural network algorithms for anomaly detection. Led development of custom web-based GIS reporting software. Improved low-level programming algorithms for data processing.
C++ / C / OpenGL / PHP / SQL / Neural Networks / Borland / Windows / Linux |
| 2005 – 2008 | Co-founder / App Developer / Instructor
SuRJE Software Solutions Inc. (self-employed, contract)
Created and published physics-based games for iOS. Technical writing, web development, Google Maps and Firebase integration and server administration. Developed course curriculum, instructed C++ programming and assessed student work with the Department of Continuing Education.
Objective-C / C++ / OpenGL / Chipmunk / PHP / JavaScript / Firebase / Google Maps / iOS / Linux |

Software Researcher / Lead Developer

2005 - 2006

PlayStarMusic Corp.

Lead developer and system administrator for an iTunes-like music store. Automated DRM encoding and video streaming. Developed a prototype for a Google-competitive search engine.

C++ / PHP / Javascript / .NET / Windows / Linux

Junior Software Engineer

2004 - 2005

Codefast.com

Ported applications to operate under a new GUI library. Developed user interfaces for software life-cycle automation.

C++ / Qt / Wx / Shell script / Windows / Unix

Web Developer / VR Artist

2002 - 2004

Nickle Arts Museum, University of Calgary (contract)

Designed and developed an interactive online catalogue for the Gerald Hushlak art exhibit with JavaScript-based features. Contributed original generative art from graduate research. Collaborated on a pioneering VR installation using 3D scanning and CAVE immersion technology.

JavaScript / C++ / OpenGL / PHP / HTML / CSS / 3D Scanning / VR

Research Assistant

2001 - 2004

VLab Computer Graphics Group / Artificial Intelligence Research Hub

Developed graphics software tools using fractal mathematics and biological modelling. Managed network setup and distributed computing infrastructure. Published five research papers and served as teaching assistant across three courses.

C++ / OpenGL / PHP / MySQL / Qt / Solaris / Unix / Linux

EDUCATION

Master's Degree

2002 - 2004

University of Calgary

Computer Science

Thesis: Agent-based Development of Natural Transportation Networks

Bachelor's Degree

1998 - 2001

University of Calgary

Computer Science

AWARDS

- Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies, 2005
- Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada, 2005
- Graduate Student Scholarship, Alberta Learning, 2004
- Best Student Paper Award (Co-author), ACAL, 2003
- CPSC Departmental Research Award, University of Calgary, 2003

PORTFOLIO



github.com: xcodegirl/career.

PUBLISHED GAMES

- Ragin' Bull - iOS, Android (2020)
- AR Picture Portal - iOS (2018)
- Fragmented Reality - iOS (2018)
- IceCream AR - iOS (2018)
- Jewel Slide - iOS, Android (2017, 2019)
- Word Slide & Find - iOS (2017)
- A Word Theme - iOS, Android (2017, 2019)
- CryptoQuip - iOS, Android (2011, 2019)
- Rafter's Loot - iOS (2011)
- Star Chase - iOS (2008)
- Bitsy - iOS (2008)